

Liver and Fatty Liver

PNPLA3, TM6SF2, MBOAT7, GCKR, HSD17B13, alpha-1 antitrypsin, iron, bilirubin (Gilbert),
gallbladder, liver enzymes and fatty liver scores and cirrhosis

Sérgio F · 02-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is
neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. Sérgio F does not have the **TM6SF2 E167K** (C/C), the second largest gene in the fatty liver. It has 1 protective allele of HSD17B13 (loss of function variable that reduces inflammation and fibrosis) and 1 allele of PNPLA3 E434K, which attenuates the effect of I148M. The genomic score of cirrhosis is at percentile 15.2, and there is no HFE C282Y, PiS, ALDH2 inactive or ABCG8 D19H. Fatty liver scores are typical (23.9, 50.6 and 78.4 percentiles; hepatic fat 65.4 and 69.4), and the load of fatty liver loci also (57.5).

Unfavourable. PNPLA3 I148M is heterozygous (C/G), the main gene of the fatty liver and fibrosis. It is added to alpha-1 antitrypsin PiMZ (SERPINA1 Z heterozygous), a fibrosis cofactor when there is hepatic fat. There is also MBOAT7 and GCKR heterozygous and HFE H63D heterozygous. The liver enzymes are genetically inclined upwards: ALT genomic score at percentile 87.0, GGT in the 94.0 (200 variants) and GGT loci load at percentile 90.7. The vesicle has mixed signals: genomic score of stones at percentile 88.8, but the 13 variant score is in the 6.3 and the loci load at percentile 34.9. It has **UGT1A1 1/28** (a little higher, benign) and ABCB11 V444A heterozygous.

What does it mean? The Sérgio F liver has two genetic cofactors of fibrosis (PNPLA3 I148M and PiMZ) partially compensated by two protectors (HSD17B13 and PNPLA3 E434K) and by the absence of TM6SF2. The basic predisposition to accumulate fat in the liver is typical. What the genes say is that if there is hepatic fat, the progression to inflammation and fibrosis is more likely than the average. Therefore, liver fat should be maintained at least, and surveillance should include fibrosis and not only enzymes.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 18 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 15 of the chip, 3 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 4 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 18 PGS Catalog scores calculated locally with 4,628,048 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 510^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 4739 loci, 4062 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

High percentile : higher risk or higher value (fatty liver, enzymes, cirrhosis, stones). The bilirubin has no judgment: slightly high values are benign and antioxidant.

Fatty liver

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Non-alcoholic fatty liver	PGS002282 (Schnurr 2022, 68 variants)	68 68 (weighted coverage 100%)	23.9	TYPICAL	typical
Non-alcoholic fatty liver	PGS000655 (Namjou 2019, 10 variants)	10 10 (weighted coverage 100%)	78.4	TYPICAL	typical
Non-alcoholic fatty liver	PGS012549 (Tang 2023, 10 variants)	10 10 (weighted coverage 100%)	50.6	TYPICAL	typical
Liver fat (MRI)	PGS002245 (Haas 2021, 8 variants)	8 8 (weighted coverage 100%)	65.4	TYPICAL	typical
Liver fat	PGS000865 (Dongiovanni 2017, 4 variants)	4 4 (weighted coverage 100%)	69.4	TYPICAL	typical

Liver enzymes

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
ALT	PGS002158 (Privé 2022, LDpred2, 774 thousand variants)	760,117 774,274 (weighted coverage 97%)	87.0	HIGH	unfavourable
ALT	PGS000816 (Pazoki 2021, 150 variants)	147 150 (weighted coverage 98%)	36.5	TYPICAL	typical
Chronically elevated ALT	PGS002732 (Vujkovic 2022, 77 variants)	75 77 (weighted coverage 86%)	31.8	TYPICAL	typical
AST	PGS002159 (Privé 2022, LDpred2, 717 thousand variants)	704,258 716,914 (weighted coverage 98%)	49.6	TYPICAL	typical
GGT	PGS002182 (Privé 2022, LDpred2, 713 thousand variants)	701,112 713,481 (weighted coverage 98%)	66.5	TYPICAL	typical
GGT	PGS000817 (Pazoki 2021, 200 variants)	193 200 (weighted coverage 95%)	94.0	HIGH	unfavourable
Alkaline phosphatase	PGS002157 (Privé 2022, LDpred2, 874 thousand variants)	857,581 874,365 (weighted coverage 97%)	70.9	TYPICAL	typical

Cirrhosis

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Cirrhosis	PGS004621 (Ghouse 2024, PRS-CS, 1.09 million variants)	1,063,635 1,089,806 (weighted coverage 97%)	15.2	LESS LIKELY	favourable
Cirrhosis	PGS000776 (Innes 2020, 9 variants)	8 9 (weighted coverage 95%)	36.5	TYPICAL	typical

Biliary vesicle

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Rocks in the vesicle	PGS012582 (Mi 2024, 13 variants)	12 13 (weighted coverage 96%)	6.3	LESS LIKELY	favourable
Vesicle stones / cholecystitis	PGS002072 (Privé 2022, LDpred2, 429 thousand variants)	421,201 428,587 (weighted coverage 98%)	88.8	MORE LIKELY	unfavourable

Bilirubin

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Total bilirubin	PGS000697 (Sinnott-Armstrong 2021, 1159 variants)	1,061 1,159 (weighted coverage 96%)	71.4	TYPICAL	without judgment
Total bilirubin	PGS002160 (Privé 2022, LDpred2, 120 thousand variants)	118,548 120,068 (weighted coverage 100%)	69.5	TYPICAL	without judgment

- **Total bilirubin:** Mildly high bilirubin (Gilbert) is associated with less cardiovascular disease; it is not a risk.

4. Key variants by topic

Variants with effect described in the literature. The judgment refers to the liver and bile ducts.

Liver fat (MASLD) and lipotoxicity

The genetic fatty liver depends mainly on the management of lipid droplets (PNPLA3), VLDL exportation (TM6SF2), phospholipid remodeling (MBOAT7) and hepatic lipogenesis (GCKR, GPAM).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PNPLA3	rs738409	I148M	C/G	chip	G (1 copy/s)	unfavourable (light)	I148M: the G (Met148) allele accumulates in the hepatocyte fat droplets and stops its mobilization. It is the main gene of the fatty liver, liver inflammation and fibrosis (OR ~1.9 by allele for MASH/fibrosis)	Strong
PNPLA3	rs2294918	E434K	A/G	imputation DR2=0.99	A (1 copy/s)	favourable	E434K: Allele A lowers the expression of PNPLA3 and attenuates the effect of I148M	Moderate
TM6SF2	rs58542926	E167K	C/C	chip	T (0 copy/s)	favourable	E167K: the T allele retains triglycerides in the liver (less VLDL exported). More hepatic fat and fibrosis, but lower LDL	Strong
MBOAT7	rs641738		C/T	chip	T (1 copy/s)	unfavourable (light)	The T allele low to MBOAT7 (remodelling of phosphatidylinositol), with more hepatic fat and fibrosis	Moderate
GCKR	rs1260326	P446L	T/C	chip	T (1 copy/s)	unfavourable (light)	P446L: T allele increases glycolysis and hepatic lipogenesis (more triglycerides and liver fat; lower fasting glucose)	Moderate
GPAM	rs2792751		C/C	imputation DR2=0.99	T (0 copy/s)	favourable	Synthesis of triglycerides in the liver (hepatic fat ↑)	Weak

Fibrosis: protective genes and cofactors

Some variants protect the liver from progression of fat to inflammation and fibrosis: HSD17B13 (loss of function) and MTARC1 A165T.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
HSD17B13	rs6834314		A/G	chip	G (1 copy/s)	favourable	Marks the insertion allele rs72613567:TA (truncated protein): protects from inflammation, fibrosis and cirrhosis (–20–40% by allele)	Moderate
MTARC1 (MARC1)	rs2642438	A165T	G/G	chip	A (0 copy/s)	neutral	A165T: A allele protects from cirrhosis and low LDL and hepatic fat	Moderate

Alpha-1 antitrypsin and iron

Alpha-1 antitrypsin Z folds poorly and accumulates in the endoplasmic reticulum of the hepatocyte. The excess iron (HFE) oxidized and gives fibrosis.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SERPINA1	rs28929474	PIZ	C/T	chip	T (1 copy/s)	unfavourable	PIZ (Glu342Lys): alpha-1 antitrypsin Z polymerizes in the reticulum of the hepatocyte. In heterozygosity (PiMZ) is a cofactor of fibrosis when there is hepatic fat (OR ~2-3 for cirrhosis)	Strong
SERPINA1	rs17580	PIs	T/T	chip	A (0 copy/s)	favourable	PIs (Glu264Val): slight deficiency; relevant combined with Z (PiSZ)	Strong
HFE	rs1800562	C282Y	G/G	chip	A (0 copy/s)	favourable	C282Y: hemochromatosis (iron overload in the liver) in homozygous	Strong
HFE	rs1799945	H63D	C/G	chip	G (1 copy/s)	unfavourable (light)	H63D: slightly ↑ iron absorption (detail in the Iron and Metals report)	Moderate

Bilirubin and conjugation (Gilbert)

UGT1A1 conjugates bilirubin and many drugs and polyphenols. *28 rises to bilirubin (benign and antioxidant).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
UGT1A1	rs887829	*28 (proxy)	C/T	chip	T (1 copy/s)	mixed	Marks UGT1A1*28 (TA7 promoter): less conjugation of bilirubin (Gilbert's syndrome in homozygous); bilirubin ↑ is antioxidant	Strong

Bilis and gallbladder

Cholesterol and bile salts transporters and biliary acid synthesis: define bile saturation and stone risk.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ABCG8	rs11887534	D19H	G/G	chip	C (0 copy/s)	favourable	D19H: C allele increases the secretion of cholesterol in the bile, with more stones in the vesicle (OR ~2 per allele)	Strong
ABCB11 (BSEP)	rs2287622	V444A	A/G	chip	G (1 copy/s)	unfavourable (light)	V444A: biliary salt pump; the allele 444A (G in strand +) lowers the expression of BSEP (collestase, hepatic injury by drugs, stones)	Moderate
CYP7A1	rs3808607		G/G	chip	G (2 copy/s)	without judgment	Biliary acid synthesis from cholesterol (promoter – 204A>C); the direction of the effect varies between studies	Weak

Alcohol (context; zero alcohol)

With zero alcohol, these genes only serve as a context.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ADH1B	rs1229984	Arg48His	C/C	imputation DR2=0.88	T (0 copy/s)	neutral	Arg48His: the T allele gives very fast oxidation of ethanol (protects from excessive consumption). Little relevant with zero alcohol	Strong
ALDH2	rs671	Glu504Lys	G/G	chip	A (0 copy/s)	favourable	Glu504Lys: inactive aldehyde dehydrogenase (very rare in the reference population)	Strong

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of each liver trait, with each allele oriented towards more risk (or higher value), compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Liver fat/fat liver	187 (183)	163 of 366	182	57.5	typical
ALT	1365 (1234)	1174 of 2468	1230	22.1	typical
GGT	1985 (1685)	1677 of 3370	1682	90.7	unfavourable
Cirrhosis/fibrosis	112 (105)	91 of 210	104	50.4	typical
Rocks in the vesicle	125 (105)	92 of 210	105	34.9	typical
Bilirubin	965 (750)	746 of 1500	749	70.6	without judgment

6. Favourable

- **TM6SF2 E167K absent** (C/C): no retention of triglycerides in the liver by this route. **Strong.**
- **HSD17B13 rs6834314 A/G**: 1 protective allele (marks the insertion of loss of function), which reduces on average inflammation, fibrosis and cirrhosis in 20–40% by allele and attenuates the effect of PNPLA3. **Moderate.**
- **PNPLA3 E434K A/G**: low expression of PNPLA3 and attenuate I148M. **Moderate.**
- **Cirrhosis**: genomic score (1.09 million variants) at percentile 15.2; typical fibrosis loci load (50.4). **Moderate.**
- **Fatty liver**: typical scores (23.9, 50.6 and 78.4 percentiles) and typical liver fat (65.4 and 69.4). **Moderate.**
- **Without HFE C282Y**, without PiS, without ALDH2 Glu504Lys and without ABCG8 D19H; GPAM C/C. **Strong.**
- **ALT**: the 150 variant score is at percentile 36.5 and the ALT loci load at percentile 22.1 (typical signals, against the genomic score). **Weak moderate.**

7. Unfavourable

- **PNPLA3 I148M C/G** (heterozygous): more fat in the hepatocyte droplets, more inflammation and more fibrosis when there is fat (OR ~1.9 by allele for fibrosis). **Strong.**
- **SERPINA1 PiZ C/T** (PiMZ): Z protein accumulates in the reticulum of the hepatocyte. In patients with liver fat, the risk of advanced fibrosis and cirrhosis is ~2–3× higher. In the lungs, it is only relevant with tobacco or tobacco. **Strong.**
- **MBOAT7 rs641738 C/T** and **GCKR P446L T/C**: 1 allele of liver fat in each. **Moderate.**
- **HFE H63D C/G**: slightly higher iron absorption. Excess iron is a fibrosis cofactor (target of ferritin < 100). **Moderate.**
- **Liver enzymes**: ALT genomics in the 87.0 and GGT percentiles in the 94.0 (200 variants; the genomics is in the 66.5), with GGT loci load at percentile 90.7. This means a genetically higher base point, which is not itself a disease. **Moderate.**
- **Vesicle**: genomic score of stones at percentile 88.8. It differs from the 13 variant score (6.3) and the loci load (34.9), so the signal is mixed. ABCB11 V444A heterozygous (biliary salt pump). **Weak moderate.**
- **UGT1A1 1/28**: mixed. The bilirubin is slightly higher (benign, antioxidant), there is less glucuronidation of some drugs and polyphenols, and there is a slight tendency for pigment stones. Strong for the genotype.

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Protein Aggregation	SERPINA1 PiZ: alpha-1 antitrypsin polymers in the hepatocyte	Unfavourable
Endoplasmic Reticulum Disorder	PiZ: reticulum stress and response to poorly folded proteins in liver	Unfavourable
Autophagy / Deregulated Macroautophagy	The elimination of Z polymers and fat droplets (lipophagy) depends on autophagy. PNPLA3 I148M hinders lipophagy	Unfavourable (mild)
Metabolic Homeostasis Decline	PNPLA3 I148M, MBOAT7 and GCKR heterozygotes (lipogenesis and retention of fat in the liver); TM6SF2 without risk	Unfavourable (mild)
Insulin resistance	GCKR P446L T/C (liver lipogenesis); typical genetic fatty liver	Neutral to unfavourable (mild)
Chronic Inflammation	Protective HSD17B13 (less hepatic inflammation) against PNPLA3 I148M	Mixed
Remodelling of the extracellular matrix	Hepatic fibrosis: PNPLA3 + PiMZ (risk) against HSD17B13 + E434K (protection); genomic cirrhosis at percentile 15.2	Mixed
Oxidative stress	HFE H63D (iron) and PNPLA3 (lipotoxicity); UGT1A1 *28 (oxidant bilirubin ↑)	Mixed
Mitochondrial dysfunction	Lipotoxicity of PNPLA3 I148M overloads the mitochondrial beta-oxidation of the hepatocyte	Unfavourable (mild)
Peroxisomal dysfunction	Peroxisomal beta-oxidation rises when there is hepatic fat (without specific variants)	Neutral

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Hepatic fat at least:** is the trigger that activates PNPLA3 and PiMZ. The main lever is to reduce visceral adiposity; maintain lean body composition and the current plan. *Suggestion (not applied).*
2. **Aerobic and strength exercise** (zone 2 + force 3×/week, as in the plan). Lowers liver fat even without changing body composition. *Suggestion (not applied).*
3. **Free sugars and liquid fructose at least** (suns, soft drinks, syrups). PNPLA3 I148M is more sensitive to fructose-induced lipogenesis and refined hydrates. High fiber (the better), as in the plan. *Suggestion (not applied).*
4. **Coffee filtered within the caffeine limit of the plan** (≤ 200 mg, only in the morning). Coffee is associated with less fibrosis and cirrhosis. *Suggestion (not applied).*
5. **Zero alcohol (already fulfills).** With PNPLA3 I148M + PiMZ is the most important rule. *Suggestion (not applied).*
6. **Fibrosis surveillance, not only of enzymes:** ALT, AST, GGT, platelets and annual FIB-4; hepatic elastography (baseline and then according to the result). Confirm PiMZ with serum alpha-1 antitrypsin and Pi phenotype. *Suggestion (not applied).*
7. **Supplements and drugs with hepatotoxic potential:** pause curcumin, luteolin, apigenin and quercetin while the ALT is high is a suggestion (not applied; the real stack has not changed). Avoid concentrated extracts of high-dose green tea, kava and anabolic steroids. Paracetamol only at a moderate dose. *Suggestion (not applied).*
8. **Sufficient** (S38 of the plan; PEMT: choline deficiency is a direct cause of fatty liver. *Suggestion (not applied).*
9. **Iron:** without iron supplements; keeping ferritin below 100 (HFE H63D; iron is a fibrosis cofactor). *Suggestion (not applied).*
10. **Zero tobacco and tobacco** (PiMZ: pulmonary protection). Vaccination against hepatitis A and B if not done. *Suggestion (not applied).*
11. **UGT1A1 *28:** inform the physician before irinotecan or protease inhibitors. Mildly high bilirubin is expected and benign. *Suggestion (not applied).*
12. **Omega-3 3.2-4 g/day (already in the plan):** low hepatic fat and triglycerides. LDL/ApoB as low as possible. *Suggestion (not applied).*

Not recommended: alcohol at any dose; iron supplements; telomerase activators. The monthly senolytic is only fisetin + quercetin and only with normal ALT.

10. Limitations

- The chip with imputation does not read rare variants of large effect (e.g. ABCB4, ATP8B1, APOB truncants, LIPA, rare variants of SERPINA1 besides Z and S) or the insertion HSD17B13 directly: a marker in strong linkage imbalance (rs6834314) was used.
- The liver enzymes scores (ALT, GGT) reflect the genetic basis point, not disease. Fat livers based on few variants are dominated by PNPLA3 and TM6SF2.
- The two stone scores in the vesicle disagree (13 variants vs genomic); the loci load is typical.
- The load of loci GWAS sums SNPs in linkage imbalance. It is a descriptive complement and does not replace the polygenic scores.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The alleles of each location are read from the Impaired VCF, which includes the SNPs of the chip. The published effect allele is compared with them; in the palindromic SNPs, strand + (GWAS Catalog and PGS Catalog) is assumed.
- **Polygenic scores:** GRCh38 files from PGS Catalog (18 scores: fatty liver, liver fat, ALT, AST, GGT, alkaline phosphatase, cirrhosis, stones in the vesicle and bilirubin). Score = Δ coefficient $\times$ dose of effect allele. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog (associations with $p \leq 5 \times 10^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** reference literature (Romeo 2008 and Trepo 2020 for PNPLA3; Kozlitin 2014 for TM6SF2; Mancin 2016 for MBOAT7; Abul-Husn 2018 for HSD17B13; Strnad 2019 for PiZ; Emdin 2020 for MTARC1, GWAS Catalog and ClinVar. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Liver fat (MASLD) and lipotoxicity	PNPLA3	rs738409	C/G	chip	G	1	strand + (palindromic; assumed +)	1 copy of risk allele
Liver fat (MASLD) and lipotoxicity	PNPLA3	rs2294918	A/G	imputation DR2=0.99	A	1	strand +	1 copy of the protective allele
Liver fat (MASLD) and lipotoxicity	TM6SF2	rs58542926	C/C	chip	T	0	strand +	without the risk allele
Liver fat (MASLD) and lipotoxicity	MBOAT7	rs641738	C/T	chip	T	1	strand +	1 copy of risk allele
Liver fat (MASLD) and lipotoxicity	GCKR	rs1260326	T/C	chip	T	1	strand +	1 copy of risk allele
Liver fat (MASLD) and lipotoxicity	GPAM	rs2792751	C/C	imputation DR2=0.99	T	0	strand +	without the risk allele
Fibrosis: protective genes and cofactors	HSD17B13	rs6834314	A/G	chip	G	1	strand +	1 copy of the protective allele
Fibrosis: protective genes and cofactors	MTARC1 (MARC1)	rs2642438	G/G	chip	A	0	strand +	without the protective allele (common genotype)
Alpha-1 antitrypsin and iron	SERPINA1	rs28929474	C/T	chip	T	1	strand +	PiMZ (Z carrier): liver fibrosis cofactor; pulmonary risk only with tobacco
Alpha-1 antitrypsin and iron	SERPINA1	rs17580	T/T	chip	A	0	strand + (palindromic; assumed +)	without the risk allele
Alpha-1 antitrypsin and iron	HFE	rs1800562	G/G	chip	A	0	strand +	without the risk allele
Alpha-1 antitrypsin and iron	HFE	rs1799945	C/G	chip	G	1	strand + (palindromic; assumed +)	1 copy of risk allele
Bilirubin and conjugation (Gilbert)	UGT1A1	rs887829	C/T	chip	T	1	strand +	1/28: a little higher bilirubin (antioxidant) and less glucuronidation of some drugs/polyphenols
Bilis and gallbladder	ABCG8	rs11887534	G/G	chip	C	0	strand + (palindromic; assumed +)	without the risk allele
Bilis and gallbladder	ABCB11 (BSEP)	rs2287622	A/G	chip	G	1	strand +	1 copy of risk allele
Bilis and gallbladder	CYP7A1	rs3808607	G/G	chip	G	2	strand +	2 copy of the effect allele
Alcohol (context; zero alcohol)	ADH1B	rs1229984	C/C	imputation DR2=0.88	T	0	strand +	without the protective allele (common genotype)
Alcohol (context; zero alcohol)	ALDH2	rs671	G/G	chip	A	0	strand +	without the risk allele